

	Since B_Q is out of the page at the centre of coil Q, B_P must be into the page at the centre. Hence direction of current in wire P flows to the right.
21.	Ans: A Since the reading on the balance increased, there must be a downward force acting on the horseshoe magnet and an upward force on the wire according to Newton's 3rd law. Using Fleming's left hand rule, since magnetic force on wire is upwards and current is from Y to X, the magnetic field must be from A (North) to the opposite face (South)
22.	Ans: A